Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/15/2006
PI Name:	Scott, Bryan B.S.		
Project Title:	Development of a Nitrifying Packed Bed Bioreactor for use in an Advanced Water Recovery System for Microgravity Environments		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	10/01/2002	**End Date:**	10/01/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
	This test addresses two challenges in microgravity wastewater reclamation for bacterial processes: nitrification and aeration. Biological wastewater processing requires sufficient nitrification, which in turn requires adequate oxygenation through aeration. This project will construct, analyze and model a gravity independent nitrification (GIN) reactor. The results of this work will be used to develop an aerobic nitrification reactor that will be coupled with an anaerobic bioreactor currently in development at the Advanced Water Recovery System Development Facility at Johnson Space Center. Performance will be measured as conversion of ammonium to nitrate as a function of residence time and ammonia load. The conversion of ammonium to nitrate is critical in maintaining the anoxic bioreactor TOC reduction efficiency. The ammonium effluent from the first stage anoxic bioreactor is fed to the second stage nitrifying reactor where the		

Task Description:	ammonium is oxidized to nitrate and then returned to the anoxic reactor as a nitrate reactant necessary for organics reduction by the anaerobic bacteria. The nitrification process is primarily accomplished by two groups of autotrophic nitrifying bacteria that can build organic molecules using energy obtained from inorganic sources, in this case ammonia or nitrite. In the first step of nitrification, ammonia-oxidizing bacteria oxidize ammonia to nitrite according to equation (1). NH3 + O2 NO2-+ 3H+ + 2e- (1) Nitrosomonas is the most frequently identified genus associated with this step, although other genera, including Nitrosococcus, and Nitrosospira, can also be present. Some subgenera, Nitrosolobus and Nitrosovibrio, can also autotrophically oxidize ammonia (Watson et al. 1981). In the second step of the process, nitrite-oxidizing bacteria oxidize nitrite to nitrate according to equation (2). NO2- + H2O NO3- + 2H+ +2e- (2) Nitrobacter is the most frequently identified genus associated with this second step, although other genera, including Nitrospina, Nitrococcus, and Nitrospira can also autotrophically oxidize nitrite (Watson et al. 1981).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	New developments in wastewater treatment are beneficial to all water purification processes in any environment.
Task Progress:	A material balance on the system predicts that only 50 mg/L of ammonium can be theoretically oxidized to nitrate given the maximum solubility of oxygen at current conditions. Given that the system is currently oxidizing 300 mg/L of ammonium the membrane oxygenator must be doubling as a bioreactor where the bacteria are consuming oxygen within the oxygenator and hence increasing the transfer of oxygen across the membrane at a rate much higher than that given by equilibrium saturation of air and water with no reaction. The Stevens reactor operated with pure oxygen at 25 psig and yet could not achieve good conversions of ammonium to nitrate at ammonium levels higher than 50 mg/L. This is probably due to poor transfer of oxygen in the silastic tubing oxygenator. A major conclusion of the results thus far is that the cross flow hollow fiber medical oxygenator provides an excellent environment for the nitrifying bacteria to thrive as well as excellent transfer of oxygen from air to the wastewater stream.
Bibliography Type:	Description: (Last Updated: 02/27/2004)